# Sampling, measurement and ring-width series

Lesson 4 · 90 minutes · From samples and records to comparable series

# Learning objectives

## Plan sampling and records

Explain a sampling plan and the metadata it needs.

## Measure widths from boundaries

Check the path, spatial calibration, boundary positions and units.

## Combine ring-width series

Distinguish detrending, tree weighting and sample depth.

# From a specimen to a chronology

01 Question and sampling plan

02 Specimens, images and records

03 Observe and measure boundaries

04 Crossdating and quality control

05 Detrending and chronology

**Return to the wood and records when a mismatch appears.**

# Start sampling with the question

## Climate

Choose species, sites and comparison areas relevant to the climate variable.

## Forest history

Plan how spatial sampling will capture disturbance and regeneration.

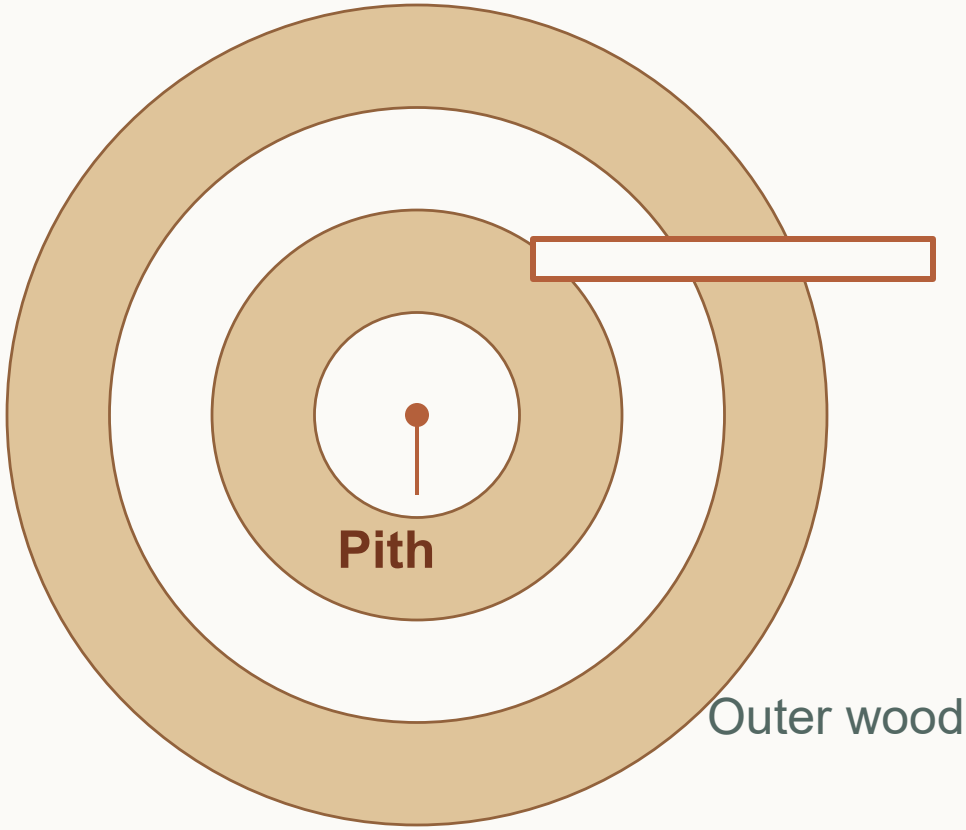
## Buildings and objects

Record member location, working or reuse, and surviving inner and outer wood.

# What material will be available?

| | |
|---|---|
| **Increment core** | **One radial record; document sampling position, losses and injuries.** |
| Disc or block | Allows comparison across radii and scars; define the sampled extent. |
| Surface image | Records the readable surface; requires a scale and imaging conditions. |

# What a core does and does not record




**Did the core reach the pith?**

Is the outermost ring preserved?

Could bending, breakage or compression affect the record?

Synthetic teaching data / schematic example, not empirical tree measurements.

# Link measurements to specimen metadata

## Identification

Link site, tree, specimen and path IDs to images and containers.

## Collection and condition

Date, species, member position, height, direction, pith, bark and losses.

## Processing and measurement

Preparation, image calibration, operator, boundaries, revisions and reasons.

# Mini exercise: plan a collection

**You want to study moisture variability in a forest with ridges and valleys.**

Choose sites, trees, radii and the records to keep.

State one limitation arising from the places you did not sample.

# Preserve order during transport and storage

## Identify immediately

Link specimen and container IDs; retain orientation and fragment order.

## Handle for the condition

Record cracks, decay and moisture; minimise distortion and mixing.

## Record the initial state

Save photographs and collection records, including missing material.

# Prepare a surface with readable boundaries

## Mount and orient

Support the core after checking its orientation and cellular structure.

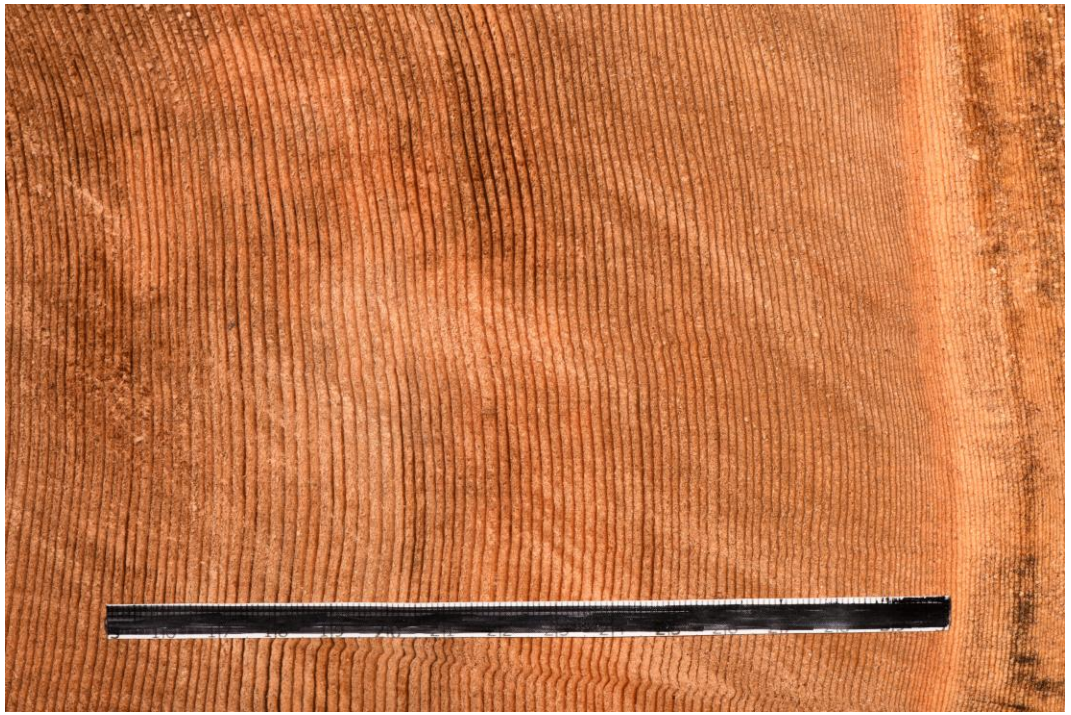
## Prepare the surface

Use preparation suited to the material to reveal boundaries and cells.

## Inspect the result

Distinguish scratches and preparation marks from narrow increments.

# Imaging a wood cross-section



Example of a wood surface. Photo: Hitoshi Yonenobu

## Distance and scale

Check imaging distance and resolution in advance; include a scale with each specimen.

## Overlapping images

Leave overlap between frames and take wider views where needed.

In the Yoshino study, adjacent images typically overlapped by one-half to two-thirds.

Yonenobu et al. (2026): Yoshino cedar

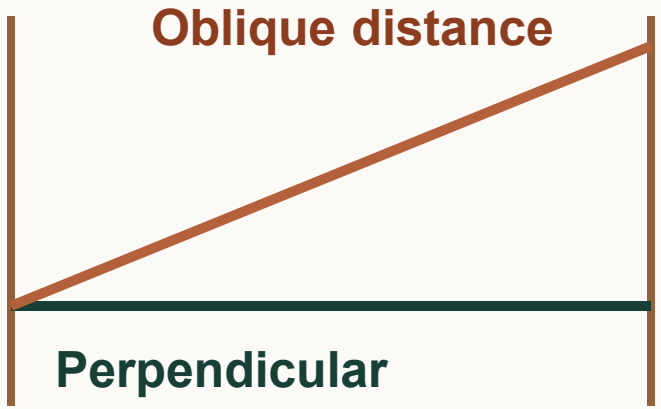
Yonenobu et al. (2026): koyamaki

# The path relative to ring boundaries

**An oblique path gives a longer distance between boundaries.**

Adjust the path to the local orientation of the boundaries.

Preserve the measured path in images and records.




Same pair of boundaries

Synthetic teaching data / schematic example, not empirical tree measurements.

# Image calibration: how many mm per pixel?

## Required accuracy: 0.01 mm / pixel spacing: 0.01 mm/pix or finer




### Calibrate using a known scale

Scale length (mm)
÷ pixels spanning that length (pix)
= pixel spacing (mm/pix)

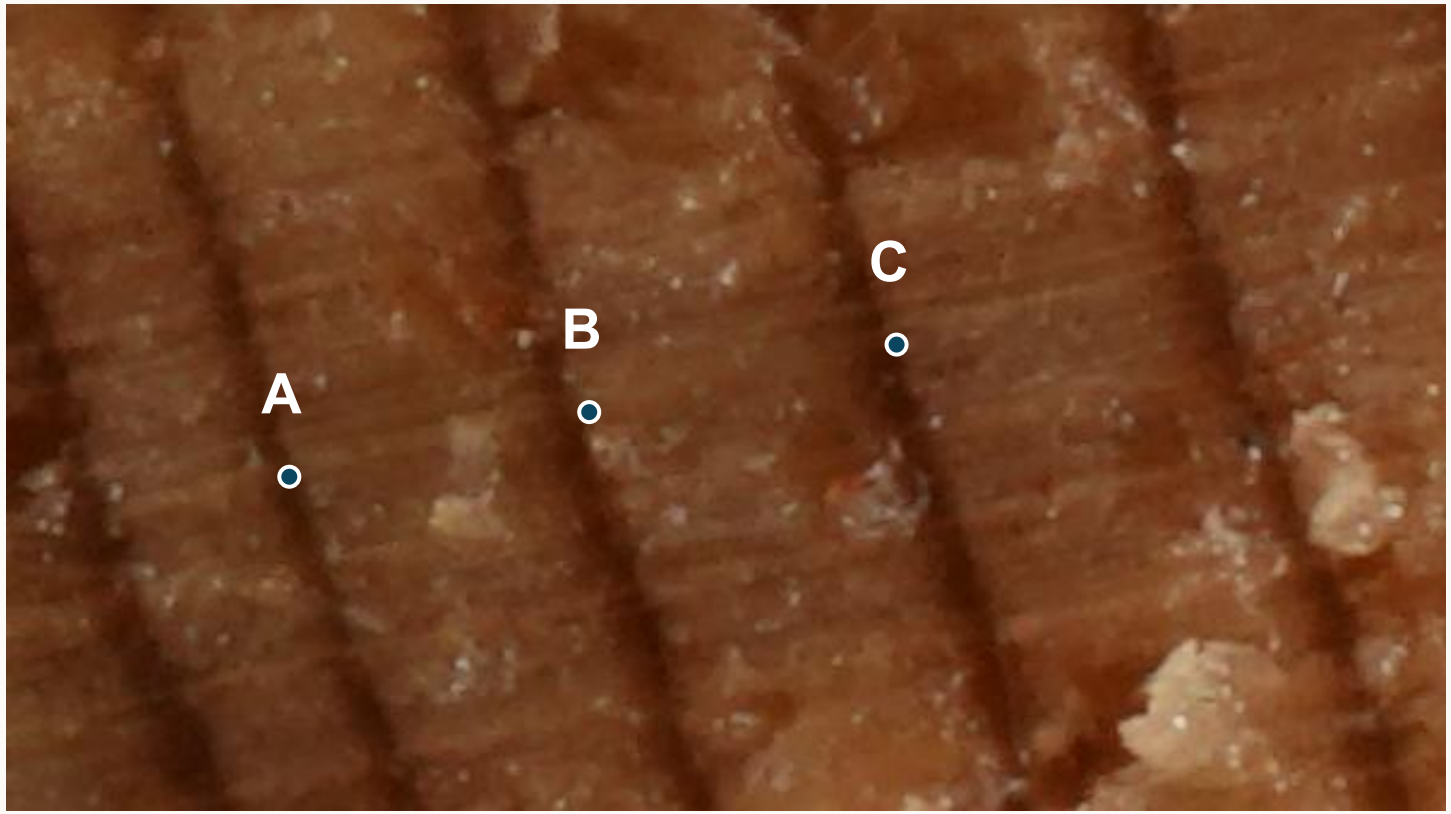
**This original: about 0.0058 mm/pix**
**Smaller values mean finer sampling.**

Pixel spacing alone does not guarantee accuracy. Check focus, scale and boundary identification.

Photo and measurements: research group / Yonenobu et al. (2026)

# Obtain widths from recorded ring boundaries

Three saved CooRecorder points overlaid on the same photograph




**A–B = 0.74 mm**
**B–C = 0.76 mm**

The rwl records 74 and 76 in units of 0.01 mm.
74 × 0.01 = 0.74 mm.

A, B and C are teaching labels, not calendar years.

Exercise A uses a separate synthetic example to practise the calculation.

Photo and measurements: research group / Yonenobu et al. (2026)

# Exercise A: calculate six ring widths

| Boundary | Position (px) |
|---|---|
| B0 | 100 |
| B1 | 160 |
| B2 | 250 |
| B3 | 340 |
| B4 | 460 |
| B5 | 520 |
| B6 | 670 |

**Scale: 0.005 mm/pix**

Record each successive pixel difference and width in mm.

Compare the sum of six widths with the endpoint distance.

Synthetic teaching data / schematic example, not empirical tree measurements.

# Exercise A: calculations and checks

| Ring | Distance (px) | Width (mm) |
|---|---|---|
| 1 | 60 | 0.30 |
| 2 | 90 | 0.45 |
| 3 | 90 | 0.45 |
| 4 | 120 | 0.60 |
| 5 | 60 | 0.30 |
| 6 | 150 | 0.75 |

**10 ÷ 2000 = 0.005 mm/pix**

(670 − 100) × 0.005 = 2.85 mm

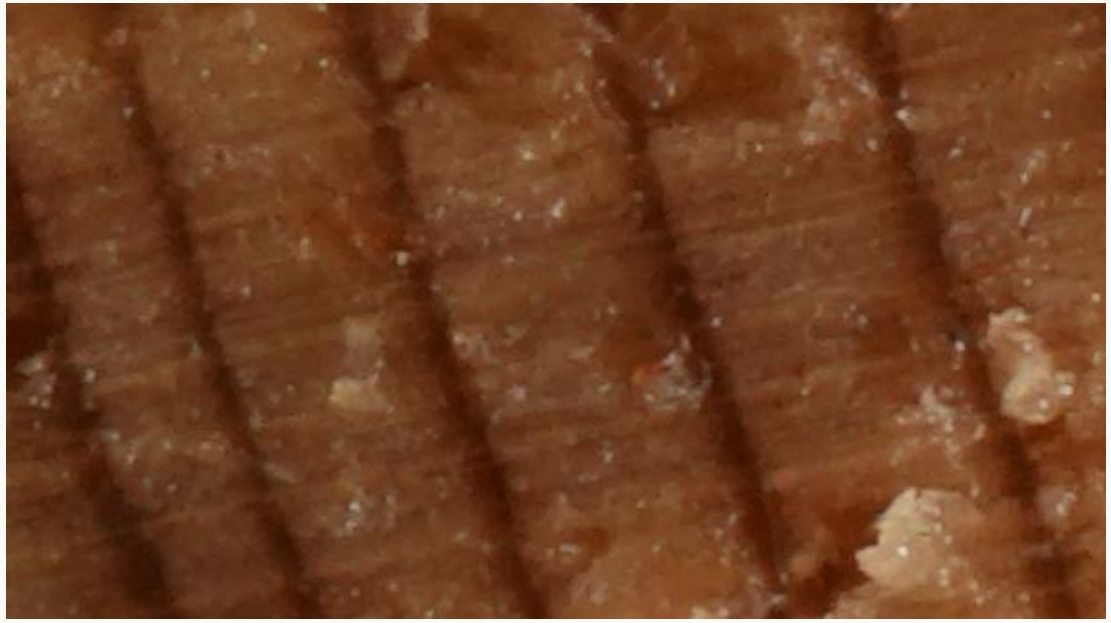
An endpoint check does not prove the boundaries were identified correctly.

Synthetic teaching data / schematic example, not empirical tree measurements.

R29 ImageJ documentation

# Pixel spacing and visibility of ring boundaries

## Crop from the original: 600 × 336 pixels



About 0.0058 mm/pix (meets spacing condition)

## Same area reduced to 150 × 84 pixels



About 0.0232 mm/pix (too coarse)

**Zooming does not add detail. After resampling an image, update its calibration.**

Photo and measurements: research group / Yonenobu et al. (2026)

# Connect measurement with dating checks

## Assign candidate calendar years

Match a dated reference and record overlap and mismatches.

## Return to the specimen

Investigate local absences, false rings, missed boundaries or measurement shifts.

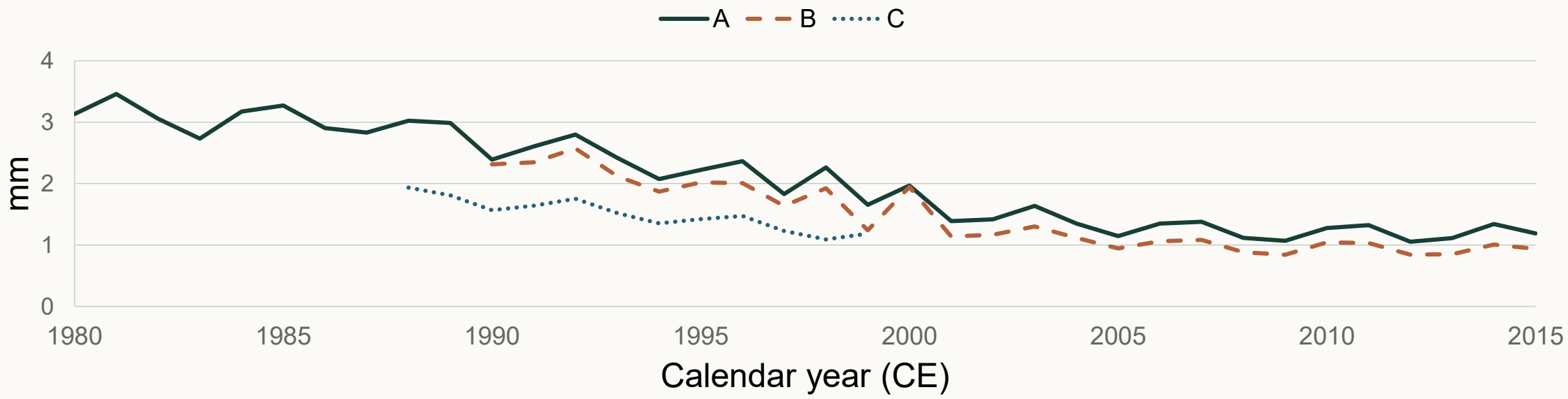
## Separate observations and interpretation

Keep missing-position and dating hypotheses separate from observations.

# Growth levels differ among trees

**Width reflects species, size, individual history and other influences.**

A raw mean alone need not isolate the signal of interest.



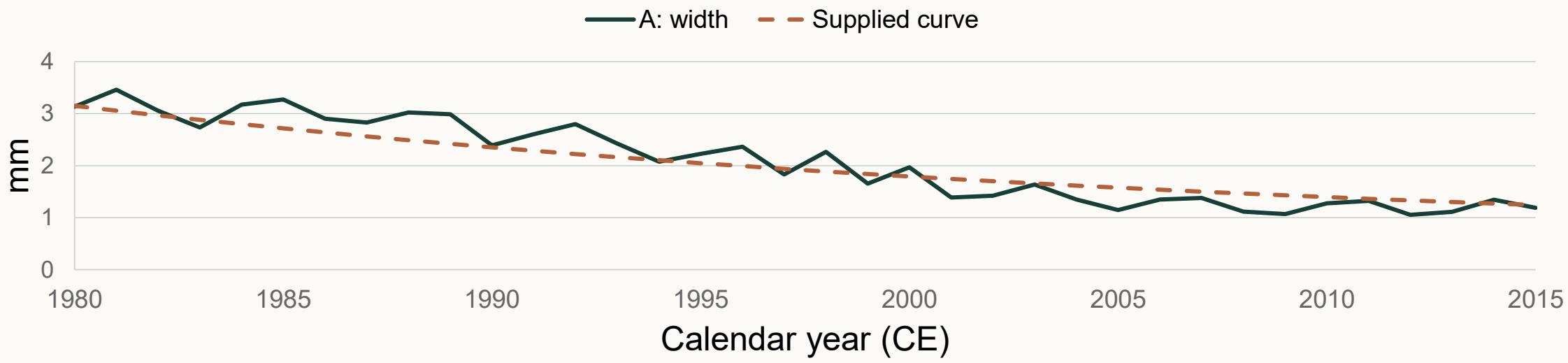

Synthetic teaching data / schematic example, not empirical tree measurements.

# Standardization relative to a growth curve

**Here, index = ring width ÷ supplied growth-curve value.**

An index of 1 matches the curve; 1.2 is 1.2 times it.

The curve choice affects the variation retained.




Synthetic teaching data / schematic example, not empirical tree measurements.

R28 dplR authors / CRAN

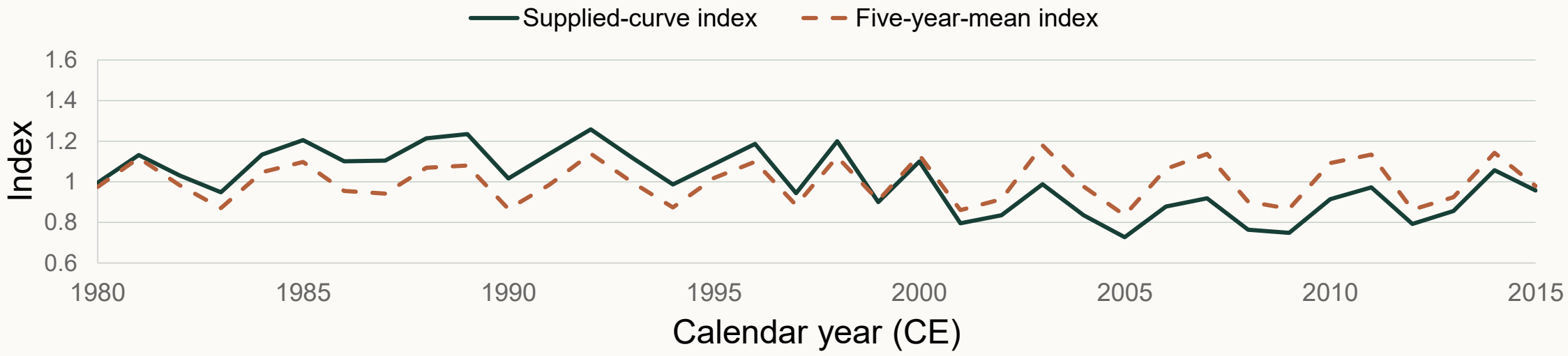
R27 NOAA NCEI

# Which variation is retained?

**A is indexed against two different reference curves.**

A flexible reference can remove longer-term variation.

Record the question, method and settings; compare results.




Synthetic teaching data / schematic example, not empirical tree measurements.

# Radii and the weighting of trees

**Tree A has two radii: 0.8 and 1.2. Tree B has one: 1.5.**

Average A first: 1.0. The equally weighted tree mean is 1.25.

A direct mean of all three radii is about 1.17, giving A more weight.

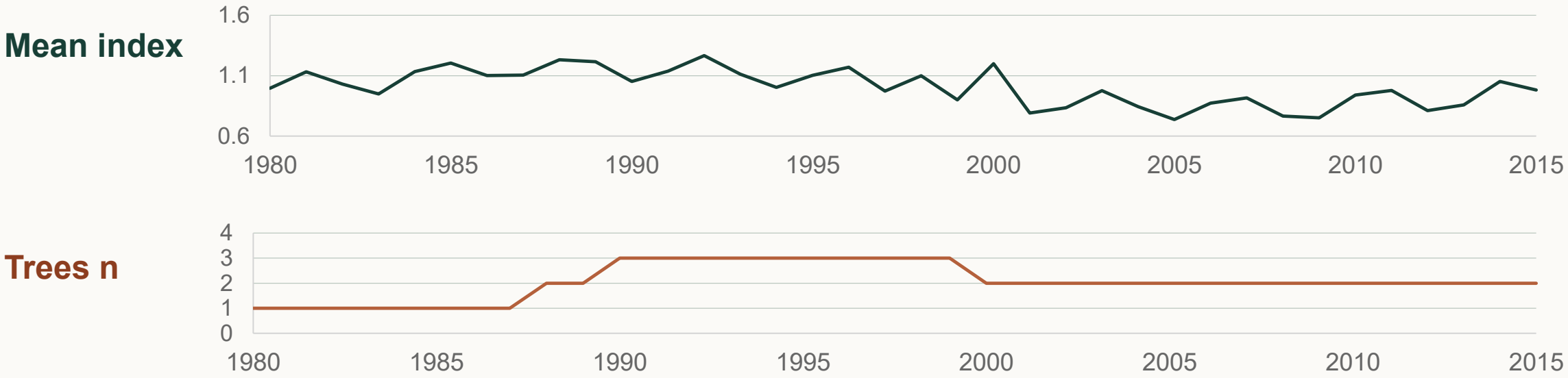
Synthetic teaching data / schematic example, not empirical tree measurements.

# Show sample depth through time

**Average the available tree indices in each calendar year.**

A continuous chronology can have changing sample depth.

Tree count alone is insufficient; consider agreement and sampling bias.




Synthetic teaching data / schematic example, not empirical tree measurements.

R27 NOAA NCEI

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# Exercise B: build indices and a chronology

**Calculate A, B and C indices from the printed handout table; no computer is needed.**

Average the available indices by calendar year and count trees n.

Compare 1998–2000 and consider the effect of replacing blanks with zero.

Synthetic teaching data / schematic example, not empirical tree measurements.

# What can the chronology establish?

## An index is not temperature or precipitation

Test environmental responses against independent evidence.

## A low index can have several causes

Distinguish drought, cold, disturbance and measurement problems.

## Review processing for the question

Removed variation may matter when studying growth release.

# Keep results traceable to the observations

**01** Specimen ID, images, boundaries, units and calibration

**02** Dating evidence and revision history

**03** Detrending settings and sample depth

**04** Preserve observations and store derived results separately.

# Check methods and evidence in research papers

## What to check

Specimen origins, methods, results and limitations.

## Use a DOI

Add https://doi.org/ before the DOI and open the link.

Example: Yoshino cedar (Yonenobu et al., 2026)

https://doi.org/10.1016/j.dendro.2026.126572

Check full-text access; see the research-paper guide for further options.

# Links for methods and further study

| | |
|---|---|
| Sample preparation and measurement | USGS Dendrochronology Lab |
| Spatial calibration in ImageJ | ImageJ documentation |
| Tree-ring data and processing | NOAA NCEI |
| dplR: detrending and chronologies | dplR authors / CRAN |
| International Tree-Ring Data Bank | NOAA NCEI |
| Glossary of Dendrochronology | WSL |

# Exit questions

**Does a matching total length prove that the boundaries are correct?**

If C is blank in 2000, should it become zero in a three-tree mean?

Write a one-sentence reason for each answer.

# Next: wood dates and human history

**Lesson 5 introduces dendroarchaeology.**

How does an outer-ring date relate to felling, working and construction?

Review surviving outer wood and evidence for the dating.